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A RENORMALIZATION GROUP STUDY OF SELF-ORGANIZED CRITICALITY

A THESIS

Presented to

The Academic Faculty

by

Jeff Hasty

in Partial Fulfillment

of the Requirements for the Degree

Doctor of Philosophy in Physics

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A RENORMALIZATION GROUP STUDY OF SELF-ORGANIZED CRITICALITY

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I dedicate this work to Sarah and Mary June Hasty for supporting me unconditionally through the duration of this work. I thank them for making my life so grand.

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SUMMARY

The purpose of this dissertation is to introduce a renormalization scheme which will enable us to clarify, improve, and test a recently proposed renormalization group (RG) framework applicable to self-organized critical systems. We present a simple 1D cellular automata model which is solvable using statistical arguments. The microstates of the system are enumerated directly, and renormalization techniques are used to analyze the model in an "exact" sense. This will provide valuable information about the structure of the technique. We use the 1D model to show where approximations must be made when applying the renormalization to the more realistic models where there is a lack of complete statistical information. We then investigate the nature of these approximations and show the conditions necessary for them to be valid. We then generalize the 1D model to include a broader class of microscopic rules. We solve for the statistical distributions for these models analytically, and generate an exact renormalization scheme. We show how each dynamical rule corresponds to an initial condition for the approximate renormalization scheme developed above, and how iteration of the map yields distribution information. The results are compared with simulations.

We then turn to two dimensions and apply the RG method to a model where exact results are known. We generate the RG map, obtain the avalanche distribution

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exponent, and compare with the known analytical value. The RG space of models generated from the directed model is then used to test further predictions of the RG picture. We develop a method for interpreting a point in the RG space as a specific cellular automata model. The model corresponding to the fixed point of the RG map is simulated, and its properties are shown to be consistent with the RG framework. We simulate other models which lie in the basin of attraction of the fixed point, and investigate their universality properties. The RG map for the two dimensional models is analyzed in detail, demonstrating how such an analysis can yield information concerning the simulations. We generate further predictions and surprises, and test them using simulations.

We conclude with a discussion of the accomplishments of this work and the questions that have been opened.

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CHAPTER I

INTRODUCTION

This chapter is devoted to the description of the phenomenology of self-organized criticality (SOC). In introducing SOC, we will present the prototypical "sand pile model" and then discuss in some detail two phenomena where the ideas have been utilized. We will next move to a discussion of the field, highlighting some of the relevant simulation, experimental and theoretical status to date. For the interested reader, the recent book by Bak [1] provides an excellent introduction to the field at the popular level. Also of note is the work of Grinstein [2], which provides an interesting (though somewhat dated) review of SOC and its relationship to nonequilibrium phenomena. We will conclude the chapter with a discussion of a recently proposed renormalization group technique applicable to SOC systems, and set the stage for the balance of this work.

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1.1 SOC - A General Description

Many phenomena can be explained theoretically by isolating a specific *scale* of interaction. This characteristic length or time scale is of great utility in that one can ignore or perturbatively treat effects which are thought to occur on other scales. For example, the nuclear interaction can be ignored if one seeks to explain some atomic phenomenon such as chemical bonding. Thus one can speak of correlations which become vanishingly small and are therefore negligible. In dynamical systems, *scale invariance* refers to the antithesis of this picture. Scale invariant phenomena have no characteristic scale of interaction; small perturbations can propagate throughout the entire system and cannot be ignored.

The physical motivation for self-organized criticality (SOC) lies in the belief that there is a large class of phenomena where scale invariance arises naturally; that is, the underlying dynamical process leads to scale invariant observables. Couched in the language of equilibrium systems, the critical state is obtained without the fine-tuning of a parameter.

The prototypical model for SOC is known as the BTW sand pile model, which was introduced nearly a decade ago in the first work on the subject [3, 4]. On an $L \times L$ lattice, this cellular automaton (CA) model is defined by two simple updating rules. First the pile is driven by randomly choosing a site (i, j) and dropping a grain

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of "sand" by updating the height $h_{i,j}$,

$$h_{i,j} \rightarrow h_{i,j} + 1 \tag{1.1}$$

Secondly, if the local height exceeds the threshold value h_c, the site relaxes by distributing sand to its nearest neighbors;

$$\begin{aligned} h_{i,j} &\rightarrow h_{i,j} - 4 \\ h_{i\pm1,j} &\rightarrow h_{i\pm1,j} + 1 \\ h_{i,j\pm1} &\rightarrow h_{i,j\pm1} + 1 \end{aligned} \tag{1.2}$$

This relaxation can then cause neighboring sites to become excited (above threshold), thus triggering a sequence of relaxation events known as an avalanche. A separation of time scales is built in to the dynamical rules; dropping does not take place until all sites are relaxed. Sand leaves the system by "falling off" at the boundary.

This model, which is intended more as a paradigm than a real simulation of sand, provides a framework for studying certain driven-diffusive systems. Independent of initial conditions, the system evolves through a transitory period to a state characterized by power laws in the statistical distributions of certain observables. For example, the avalanche size is defined as the number of sites which relax in a single avalanche, with distribution given by

$$D(s) \sim s^{1-\tau}, \tag{1.3}$$

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where $D(s)$ is the probability that s or more sites are involved.

There are of course numerous spatial and temporal measurements which can be made, and there has been considerable work in the direction of deducing various scaling laws which relate their distribution exponents [5]. We will discuss such issues below, but for now we will maintain a narrow focus, concentrating solely on the avalanche size distribution.

1.2 Some Pedagogical Examples

1.2.1 Earthquakes

SOC has been invoked to explain many diverse phenomena, with earthquakes [6, 7, 8, 9] being perhaps the most widely cited examples. An empirical result describing the distribution of earthquake sizes is the well established Gutenberg-Richter law [10]. In figure (1.1) the earthquake size distribution obtained from '73-'83 along the New Madrid fault zone in the central United States is plotted [11]. The distribution obeys $E(m) \sim m^{-\beta}$, where $E(m)$ is the number of earthquakes of energy m or greater. That such a simple fit would arise is remarkable, but perhaps even more astounding is the claim that this distribution seems to be obeyed in any geographical region, with the only region-specific quantity being the value of the exponent β.

A simple model used by geophysicists to describe one class of earthquakes

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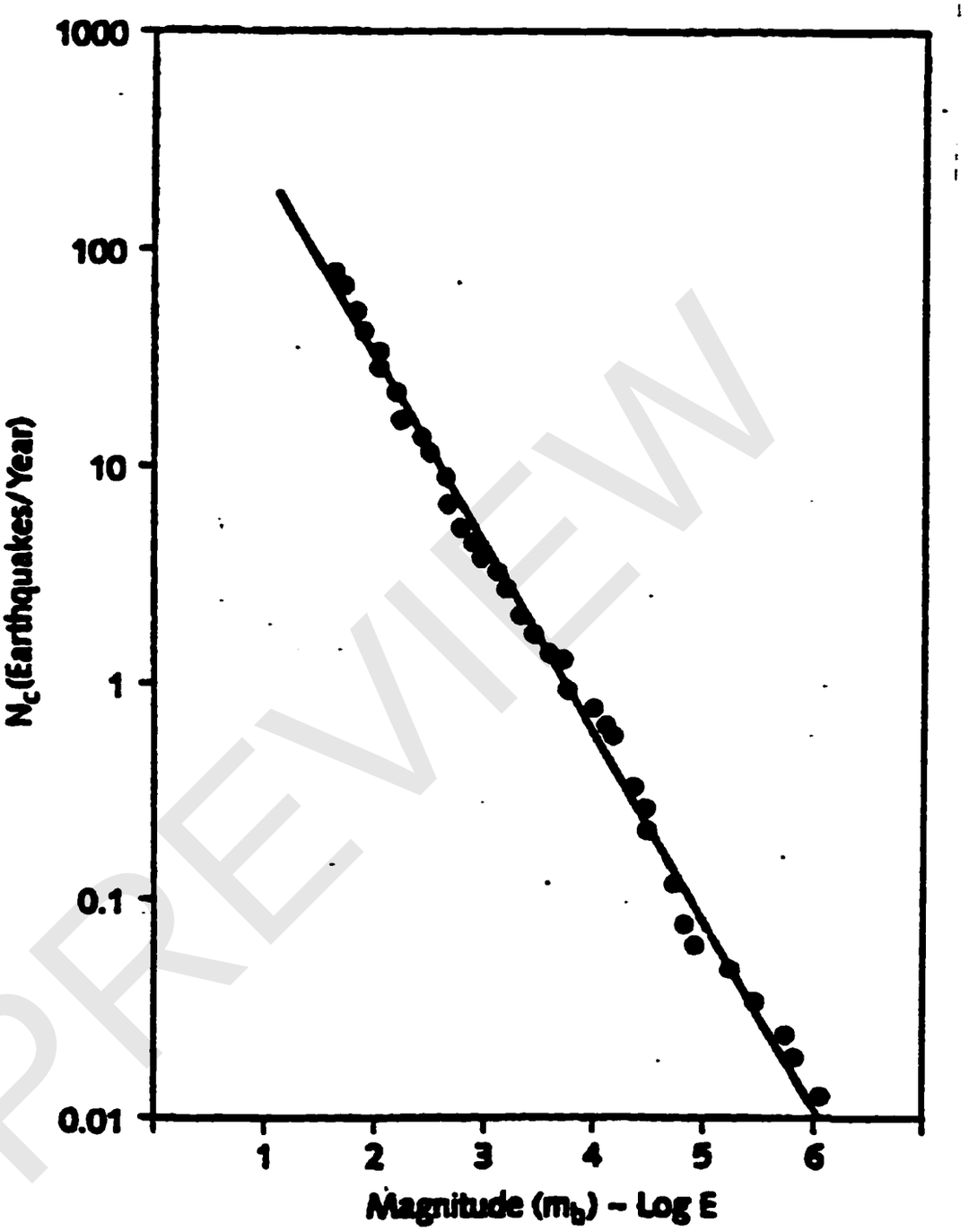


Figure 1.1: Earthquake size distribution along the New Madrid fault from 1974-1983, reproduced from [1]

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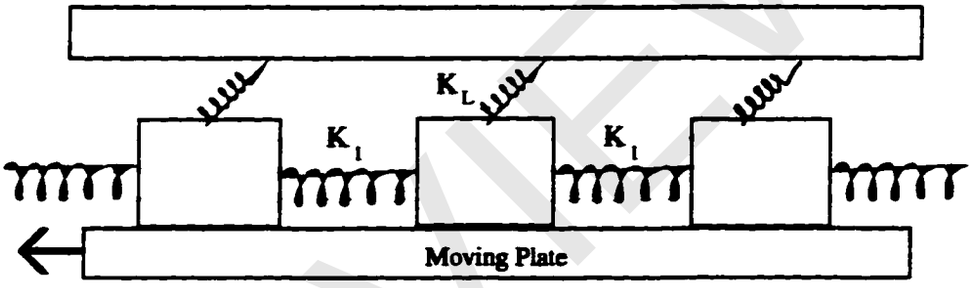


Figure 1.2: A side-view schematic of the Burridge-Knopoff spring-block model. The two dimensional model is isotropic in the unshown direction.

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